TURKEY VULTURE SEX CLASSIFICATION USING MORPHOMETRIC MEASUREMENTS AND MACHINE LEARNING

Jayana Sarma

GUIDED FEEDBACK QUESTIONS

- Is my explanation of the final model choice, especially the comparison between the 9-variable and 3-variable logistic models, clear and scientifically convincing?

- Are my results visuals and tables easy to follow during the presentation, or do any of them need simpler explanation or clearer emphasis on the key takeaway?

- Does the presentation flow logically from motivation to methods to results and conclusion, or are there transitions that should be improved?

PRESENTATION ROADMAP

- Background and motivation
- Study data and modeling approach
- Results and model comparison
- Limitations and future directions

INTRODUCTION: WHAT I STUDIED AND WHY

- Turkey Vultures show little obvious external sexual dimorphism
- Field sex identification is difficult
- Sex-specific data matter for ecology, movement, breeding, and population studies

STATEMENT OF PROBLEM AND SOLUTION

Problem

- Can morphometric measurements predict sex in *Cathartes aura meridionalis* with accuracy sufficient to be useful for field research?
- Which morphometric measurements contribute most strongly to sex prediction in this population?
- Which statistical or machine-learning model performs best for this classification task?

Solution

- Built a region-specific classification framework using western Montana birds with laboratory-confirmed sex
- Compared logistic regression, LDA, random forest, and regularized regression
- Evaluated whether a simpler reduced logistic model could retain strong performance and be more practical for field use

BRIEF LITERATURE REVIEW

- **Griffiths et al. (1998):** Molecular sexing is highly reliable, but it requires laboratory processing and is less practical for immediate field use.
- **Gaby (1982):** Morphometric measurements showed that sex separation may be possible in Turkey Vultures, at least in a Florida population.
- **Dechaume-Moncharmont et al. (2011):** Morphometric sexing can be useful, but performance depends heavily on variable choice, validation, and overlap between sexes.
- **Guo et al. (2024):** Regional variation in body form means morphometric rules may not transfer well across populations, so local validation is important.

MODELING DATASET AND PREDICTOR SET

- **Data source:** Raptor View Research Institute (RVRI)
- **Study population:** Turkey Vultures captured in western Montana
- **Field data collected during capture:** morphometric measurements
- **Sex source:** laboratory-confirmed sex from blood samples using molecular sexing
- 77 total captures; 65 birds with laboratory-confirmed sex used for modeling
- 21 morphometric predictors retained
- Missing numeric values handled with median imputation
- Models compared using repeated 5-fold cross-validation (10 repeats)
- **Main evaluation metric:** AUC; sensitivity and specificity also reviewed

METHODS: WORKFLOW

- RVRI capture data

  → Lab-confirmed sex subset selected

  → Morphometric predictors cleaned

  → Median imputation

  → Candidate model comparison

  → Reduced logistic model comparison

  → Final practical model

MODELING AND EVALUATION

Candidate models

- Logistic regression
- LDA
- Random Forest
- Regularized regression

Evaluation

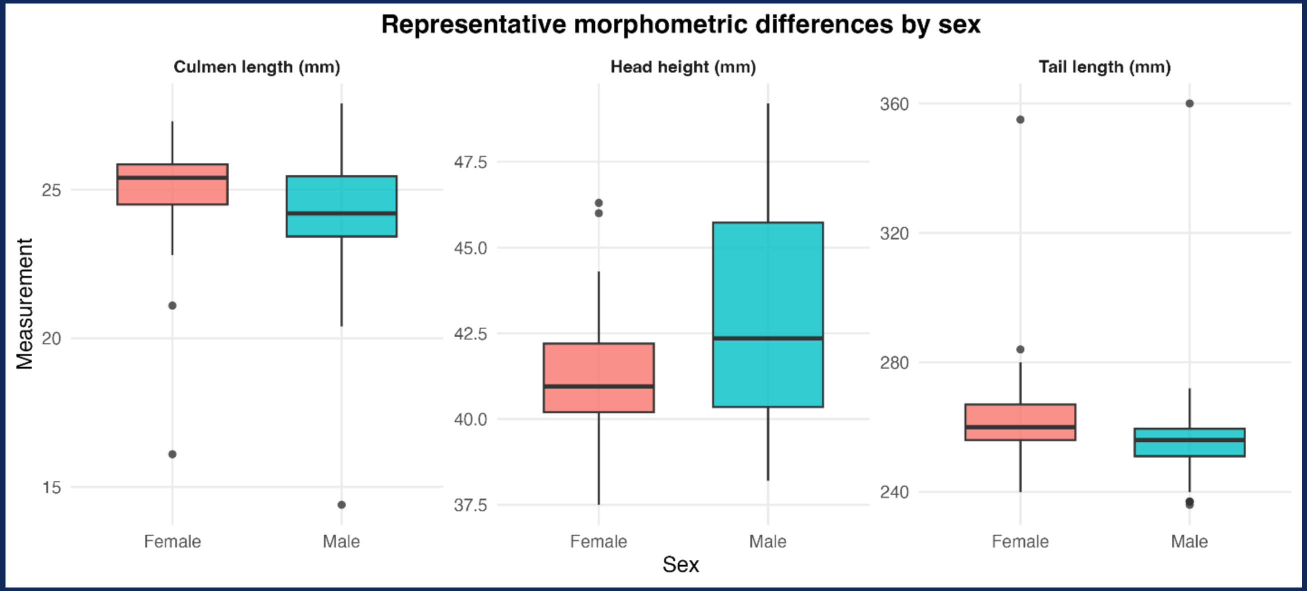
- Repeated 5-fold CV
- 10 repeats
- AUC
- Sensitivity / Specificity

Logistic comparison

- 9-variable
- 5-variable
- 3-variable

RESULTS: MALES AND FEMALES OVERLAPPED STRONGLY IN RAW MEASUREMENTS

- Male and female Turkey Vultures showed **substantial overlap** across most measurements
- No single morphometric variable provided **complete separation**
- Some variables still showed modest differences in central tendency
- This pattern suggested that **multivariable classification** would be more appropriate than one-variable thresholds



STEPWISE LOGISTIC REGRESSION PERFORMED BEST OVERALL

I first compared **four candidate model families**

- Stepwise logistic regression
- Random forest
- Stepwise LDA
- Regularized regression

Stepwise logistic regression had the strongest cross-validated discrimination

Full-data AUC values were higher than cross-validated AUC values for all models

This showed that in-sample performance was **optimistic**, so model choice should be based on cross-validation

Model	Cross-validated AUC	Full-data AUC
Stepwise logistic regression	0.807	0.930
Random Forest	0.782	1.000
Stepwise LDA	0.762	0.884
Regularized regression	0.644	0.898

RESULTS: INCLUDING AGE REDUCED PREDICTIVE PERFORMANCE

- Age was evaluated as an additional candidate predictor
- Including age did **not** improve classification performance

Mean cross-validated AUC:

Without age: 0.807

With age: 0.699

- Because age lowered predictive performance and added complexity, it was **not retained**

Model	Mean Cross-Validated AUC
Stepwise logistic regression without age	0.807
Stepwise logistic regression with age	0.699

RESULTS: COMPARING LOGISTIC MODELS OF DIFFERENT COMPLEXITY

After identifying logistic regression as the strongest model family, I compared:

- a **9-variable stepwise logistic regression model**
- a **5-variable reduced logistic regression model**
- a **3-variable reduced logistic regression model**

The **9-variable model** achieved the highest mean CV AUC

The **3-variable model remained competitive**

The **5-variable model did not improve on the 3-variable model**

Logistic model	Number of predictors	Mean CV AUC	95% CI for Mean AUC	Sensitivity	Specificity
9-variable stepwise logistic regression	9	0.807	[0.776, 0.837]	0.788	0.712
5-variable reduced logistic regression	5	0.735	[0.700, 0.769]	0.750	0.604
3-variable reduced logistic regression	3	0.742	[0.706, 0.779]	0.775	0.580

RESULTS: WHY THE 3-VARIABLE MODEL WAS THE MOST DEFENSIBLE PRACTICAL MODEL

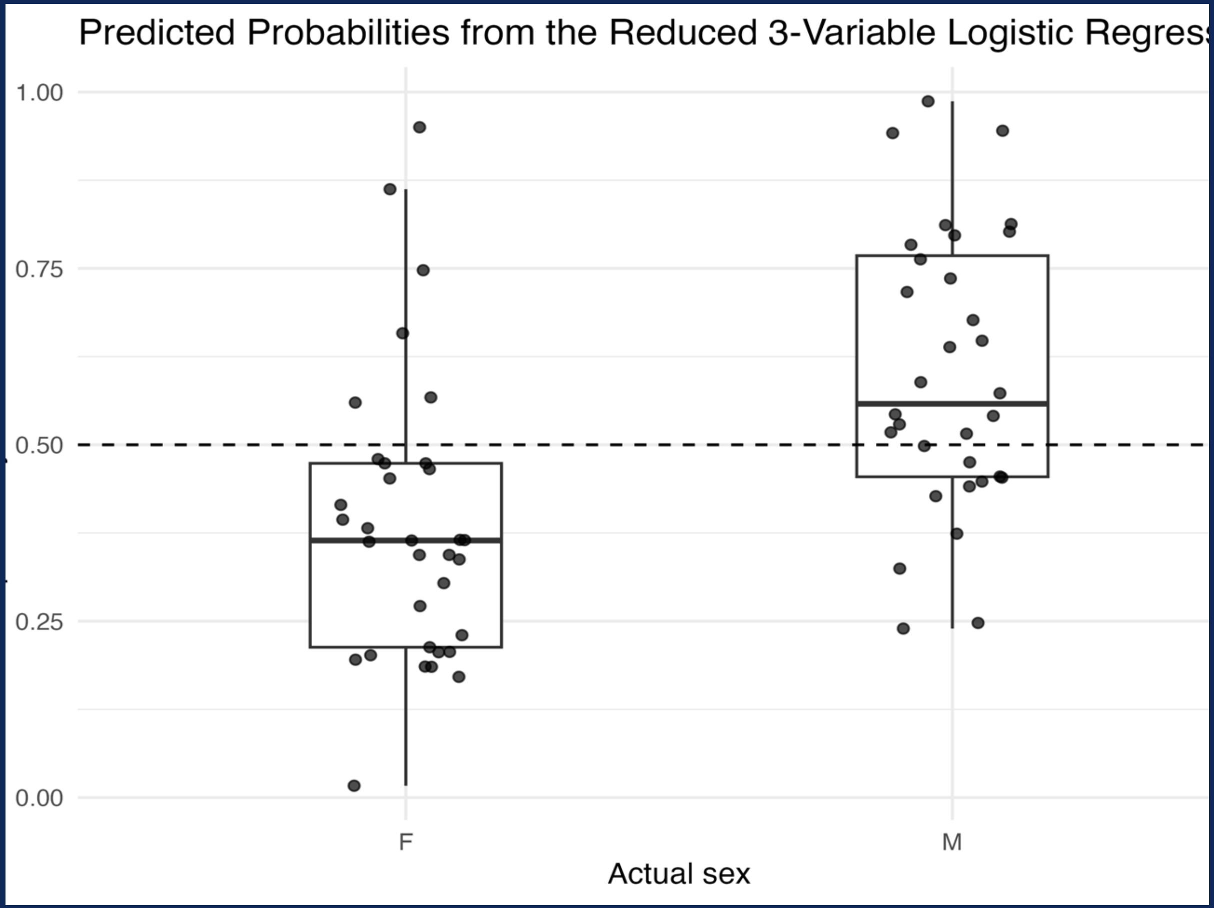
Why this model was preferred

- The 9-variable model came from stepwise AIC; the 5- and 3-variable models were then evaluated as reduced logistic models.
- Competitive cross-validated performance
- Fewer measurements required in the field
- Clearer interpretation than the 9-variable model
- All three predictors had odds-ratio 95% confidence intervals excluding 1.00

Predictor	Coefficient	Odds Ratio	95% CI for OR	p-value
Tail length (mm)	-0.051	0.951	[0.909, 0.994]	0.025
Culmen length (mm)	-0.514	0.598	[0.404, 0.884]	0.010
Head height (mm)	0.291	1.337	[1.033, 1.731]	0.027

RESULTS: THE FINAL MODEL SHOULD BE INTERPRETED PROBABILISTICALLY

- The reduced model outputs a **probability of male classification**
- Many birds were assigned probabilities well above or below **0.50**
- Birds near **0.50** reflected greater classification uncertainty
- The model is best used as a **probabilistic screening tool**, not a definitive replacement for molecular sexing



RESULTS: WHY I DID NOT AUTOMATICALLY CHOOSE THE 9-VARIABLE MODEL

Predictor	Odds Ratio	95% CI for OR
Wing chord (mm)	0.962	[0.918, 1.008]
Wingspan (mm)	0.992	[0.985, 1.000]
Tail length (mm)	0.963	[0.924, 1.004]
Hallux length (mm)	0.423	[0.182, 0.980]
Tibia length (mm)	0.639	[0.463, 0.882]
Culmen length (mm)	0.737	[0.484, 1.122]
Culmen width (mm)	0.425	[0.203, 0.889]
Head width (mm)	0.549	[0.276, 1.090]
Head height (mm)	1.759	[1.168, 2.647]

- The 9-variable model had the highest mean cross-validated AUC
- However, several adjusted odds-ratio confidence intervals included **1.00**
- This indicated that some predictor effects were estimated imprecisely
- The added complexity therefore did not clearly justify preferring the larger model for future prediction

LIMITATIONS

Small sample size
Only 65 birds had laboratory-confirmed sex, which limits model stability.

Strong overlap between sexes
Male and female morphometric measurements were not clearly separated.

Missing data required imputation
Some predictors had missing values, which adds uncertainty.

Region-specific dataset
The model was built using western Montana birds and may not transfer directly to other populations.

Not a replacement for molecular sexing
Morphometric classification is useful as a screening tool, but not as a definitive method.

FUTURE WORK

Validate the model on new birds
Test the reduced model on an independent dataset.

Test the model across years and observers
Check whether results stay stable under different field conditions.

Compare with additional simple field-usable models
Explore whether other small predictor sets perform similarly.

Evaluate use in other regional populations
Test whether the model transfers beyond western Montana.

REFERENCES

Dechaume-Moncharmont, F.-X., Monceau, K., & Cézilly, F. (2011). *Sexing birds using discriminant function analysis: A critical appraisal.* **The Auk, 128**(1), 78–86.

Gaby, S. R. P. (1982). *Age-specific resource utilization by turkey vultures in Florida.* **The Journal of Wildlife Management, 46**(2), 473–480.

Gill, F. B. (2007). *Ornithology* (3rd ed.). W. H. Freeman.

Griffiths, R., Double, M. C., Orr, K., & Dawson, R. J. G. (1998). *A DNA test to sex most birds.* **Molecular Ecology, 7**(8), 1071–1075.

Guo, K., Huang, Y., Sun, N., Ren, Y., Luo, D., & Guo, Z. (2024). *Re-testing Bergmann's rule in two species of passerines wintering in China.* **Journal of Avian Biology, 2024**(4), e03274.

Kirk, D. A., & Mossman, M. J. (1998). *Turkey vulture (Cathartes aura).* In A. Poole & F. Gill (Eds.), **The Birds of North America** (No. 339).

THANK YOU